Ghislain THEUBO

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Professional Summary

Embedded-oriented Software Engineer (Ensimag) with hands-on **C/FreeRTOS** real-time control at **INRIA** (assistive wheelchair — sensors, latency, user validation) and industrial software discipline from Dassault Systèmes and Gustave Roussy. Strong in C/C++, comfortable with Python, multi-threaded systems, Git, and rigorous coding standards. Seeking **Embedded Software Engineer** at Harmattan AI (Paris, on-site) to contribute to UAV Embedded Linux/RTOS software, sensor drivers, telemetry, and field-ready reliability.

Core Strengths

C/C++ · **FreeRTOS / real-time** · Sensors & drivers · Telemetry pipelines · Debugging under constraints · Python plus · Architecture & reliability · Cross-team collaboration · English · Mission ownership

Experience

Shift Technology – Platform Software Engineering — Paris, France
Software Engineer (CDI – Platform / Backend) — *Sept 2026 – Nov 2026*

- Contributed to the **platform / foundations** squad building shared services used by product teams: design and implementation of **C# / .NET** components and **high-performance APIs** on **Microsoft Azure**.
- Worked with cloud data and messaging building blocks (**Azure Cosmos DB**, **Service Bus**), configuration/transversal services, and infrastructure-as-code awareness (**Terraform**) in a hybrid SaaS environment.
- Practiced production engineering habits: design discussions, code reviews, monitoring mindset, incremental delivery, and collaboration with platform and product engineers.

Gustave Roussy & CentraleSupélec MICS Lab — Paris / Saclay, France
Software Engineer (CDD – Backend / Data Platforms) — *Jan 2026 – Sept 2026*

- Production backends (Python/FastAPI) for clinical/research workflows: validation, RBAC, structured logging; PostgreSQL tuning (∼**30%** faster critical reads).
- Docker + GitHub Actions; documentation for operational handoff; translated ambiguous domain needs into reliable software.

Dassault Systèmes – 3DEXPERIENCE / 3DSwym R&D — Vélizy-Villacoublay, France
Software Engineer Intern (Full-Stack Module Rewrite) — *Mar 2025 – Sept 2025*

- Full-stack overhaul of the legacy **Survey module**: Java 17 / Spring Boot / Hibernate, TDD, design patterns.
- Results: ∼**35%** performance improvement and ∼**85%** test coverage — maintainable product delivery in a large R&D org.

INRIA Grenoble – Humanlab / Froll'N'Roll — Montbonnot-Saint-Martin, France
Research Intern – Real-Time Control Modules (Assistive Wheelchair) — *May 2024 – Aug 2024*

- Developed real-time control modules in **C/FreeRTOS** and Python for an electrically assisted manual wheelchair — continuous validation with the end user.
- Redesigned communication/data pipelines under latency constraints; improved responsiveness by ∼**40%**; open/documented research codebases.
- Human-centered engineering: DIY hardware/software integration, iterative prototyping, multidisciplinary SED team.
- Closest analog to Harmattan: embedded RTOS + sensor pipelines + collaboration with algorithmic/hardware partners.

LIG – Laboratoire d'Informatique de Grenoble — Grenoble, France
Cloud Infrastructure Intern (VM Migration on OpenStack / Grid'5000) — *Feb 2025 – Mar 2025*

- Automated VM migration experiments on OpenStack/Grid'5000; reproducible scripts and performance/availability analysis.

Selected Projects

INRIA Froll'N'Roll – Assisted Wheelchair Control | C, FreeRTOS, Python — INRIA

- Real-time control on embedded constraints; sensor processing; +40% responsiveness; continuous end-user validation.

UAV-style Sensor Driver Sketch | C/C++, UART/SPI/I2C notions, datasheets — GitHub

- Datasheet-driven driver interfaces, RTOS task design notes, and bring-up checklist — ramp for Harmattan sensor integration.

Telemetry Ring Buffer Lab | C++, Catch2 — GitHub

- Bounded buffers, drop policies, and unit tests for high-rate telemetry paths — quality bar for constrained environments.

Technical Skills

Embedded: C, C++, FreeRTOS, multi-thread/process notions, firmware architecture basics, Git, build systems
Plus: Python, Linux, latency-aware debugging, serial protocols (UART/SPI/I2C notions)
Domain signal: INRIA assistive robotics / real-time; eagerness for Embedded Linux/RTOS UAV work
Languages: English professional; French fluent

Education & Certifications

Grenoble INP - Ensimag — Grenoble, France
Engineering Degree – Information Systems / Software Engineering (Double Degree) — *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé — Yaoundé, Cameroon
Master's Degree in Computer Engineering – Top 12% of class — *2019 – 2023*

Certifications: Programming Foundations – TDD (LinkedIn, 2025) · Primerli – Insurance Industry Foundations (2025)

Languages & Extra

English: bilingual / professional · **French:** fluent · **Spanish:** intermediate
Extra: academic tutoring (UGA DAEU maths; mentoring) · tennis, football